Data Analytics with Python and Data Storage Reflection

For this project I used the UK Government Archived Price Paid Data: 1995 to 2017 (HM Land Registry, 2025), selecting a subset from 2013–2017. This data is structured as CSV. To complete the task I used ChatGPT to create the code (OpenAI, 2026a) (OpenAI, 2026b), in line with industry expectations (AIM Media House, 2025).

Whilst this dataset is not SQL, it is structured data from a single, consistent data source and was stored locally. The dataset was small: 4,921,437 rows and 16 columns before it was cleaned. However, there was still a noticeable lag in the retrieval of data. This highlights how important the optimisation of data structure can be.

There was also an unexpected duplication of data. The first cleaning algorithm produced unexpected results and the analysis suggested some properties were sold more than 600 times in four years. When I verified this, it was clear that the full address was not being considered and properties, like blocks of flats, were being identified as the same property. This also highlighted the need for human verification and quality control of data generated from AI.

Moving forward, it would be interesting to conduct an experiment to convert the data to SQL and NoSQL formats and compare data retrieval time and to see if this improves the quality of the data grouping, resulting in less manual “cleaning”.

Due to the small dataset size and high consistency and structure, SQL would likely be sufficient and potentially faster due to indexing. However, NoSQL could be more effective for real-time data analysis, if the dataset grows substantially, or if there is a need to combine data from multiple different sources with an inconsistent data structure.

References

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